ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C5 · SIMPLE MACHINES

Name the machine

★☆☆
Grade 3

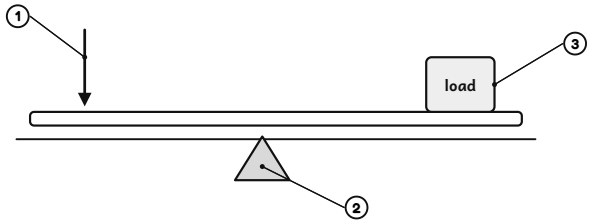
Learning goals

- Label a lever.
- Name simple machines.

1 Label the lever. [3]

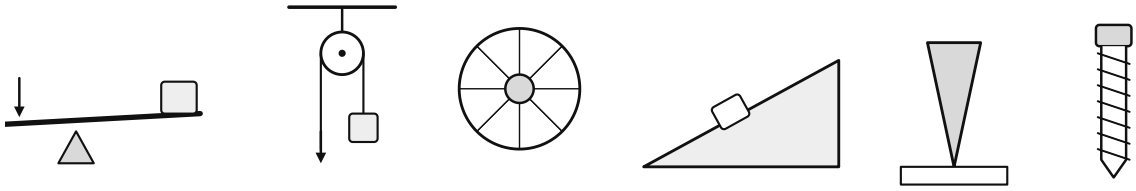
LABELS

fulcrum (pivot) load effort (push)



(1) effort (push) (2) fulcrum (pivot) (3) load

2 Name each simple machine. [6]



A	B	C	D	E	F
lever	pulley	wheel and axle	inclined plane	wedge	screw

3 Draw a line to match. [4]

Draw a line to match each pair, or write the letter.

scissors ●	→ A	● **A** lever
wheelchair slope ●	→ C	● **B** pulley
flag pole rope ●	→ B	● **C** ramp (inclined plane)
bicycle chain wheels ●	→ D	● **D** gears

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C5 · SIMPLE MACHINES

How do machines help?

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Grade 3

Learning goals

- Explain how machines make work easier.

1 **Why is it easier to open a tin lid with a long spoon than with a coin?** [2]

The spoon is a longer lever, so you need less force.

2 **Why do roads up steep hills zig-zag?** [2]

The zig-zag makes a longer, gentler ramp, so cars need less force to climb.

3 **How does a pulley help someone raise a flag?** [2]

They pull down on the rope and the flag goes up; pulling down is easier.

4 **Draw a lever you use at home. Mark the fulcrum, effort and load.** [3]

My lever

e.g. scissors, bottle opener or spoon, with fulcrum, effort and load marked.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · C5 · SIMPLE MACHINES

Does a ramp make it easier?

★★☆
Grade 3

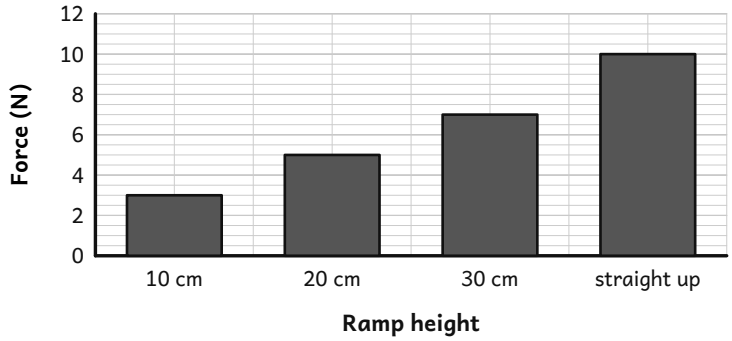
Learning goals

- Measure force in newtons.
- Draw a bar chart and conclude.

Class 3C used a forcemeter to pull a toy truck up ramps of different heights. Lifting it straight up needed 10 N.

1 **Draw a bar chart.** [3]

Ramp height	10 cm	20 cm	30 cm	straight up
Force (N)	3	5	7	10



2 **Which ramp needed the least force?** [1]

the lowest one (10 cm)

3 **Complete: the lower (gentler) the ramp, the ___ force is needed.** [1]

less

4 **What did they keep the same?** [1]

the same truck, the same ramp length, the same forcemeter

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C5 · SIMPLE MACHINES

From dheki to rice mill

★★★
Grade 3

Learning goals

- See how machines change daily life.

Science | History | Work

For hundreds of years, village women in Bangladesh husked rice with a **dheki**. One person stepped on the end of a long wooden beam. The beam turned on a pivot and the other end lifted, then thumped down onto the rice to knock off the husks. It was slow, hard work.

Today most rice is husked in **rice mills**. A motor turns **gears** and **belts** that work the machines, and one mill can husk tonnes of rice in a day. Some families still use a dheki to make special rice flour for winter pitha.

Based on rice husking in Bangladesh.

1 **What kind of simple machine is a dheki?** [1]

a lever

2 **How did the dheki make work easier?** [2]

Pressing the long end with your foot lifted the heavy beam; the lever made it easier.

3 **Write one good thing and one problem about rice mills.** [2]

Good: much faster, less hard work. Problem: needs electricity or fuel; some people lost their work.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C5 · SIMPLE MACHINES

Show what you know

★★★
Grade 3

Learning goals

- Use machine words.

1 Find the words.

N	G	M	L	E	V	E	R	H
L	E	A	C	H	L	Y	G	H
P	A	E	F	F	O	R	T	T
D	R	I	R	A	K	P	K	V
P	U	L	L	E	Y	H	N	F
K	W	W	H	E	E	L	R	G
L	O	A	D	W	Y	R	A	F
D	Y	G	P	S	V	L	U	S
H	F	F	B	V	R	A	M	P

effort gear lever load pulley ramp wheel

2 The point a lever turns on is the... [1]

○ load
● **fulcrum**
○ gear

3 A ramp lets you use... [1]

○ more force
● **less force**
○ no force

4 Challenge. Design a machine to lift a bucket of sand to a first-floor window. Label the simple machines you used. [3]

My machine

e.g. a pulley fixed above the window with a rope, maybe with a winding handle (wheel and axle).